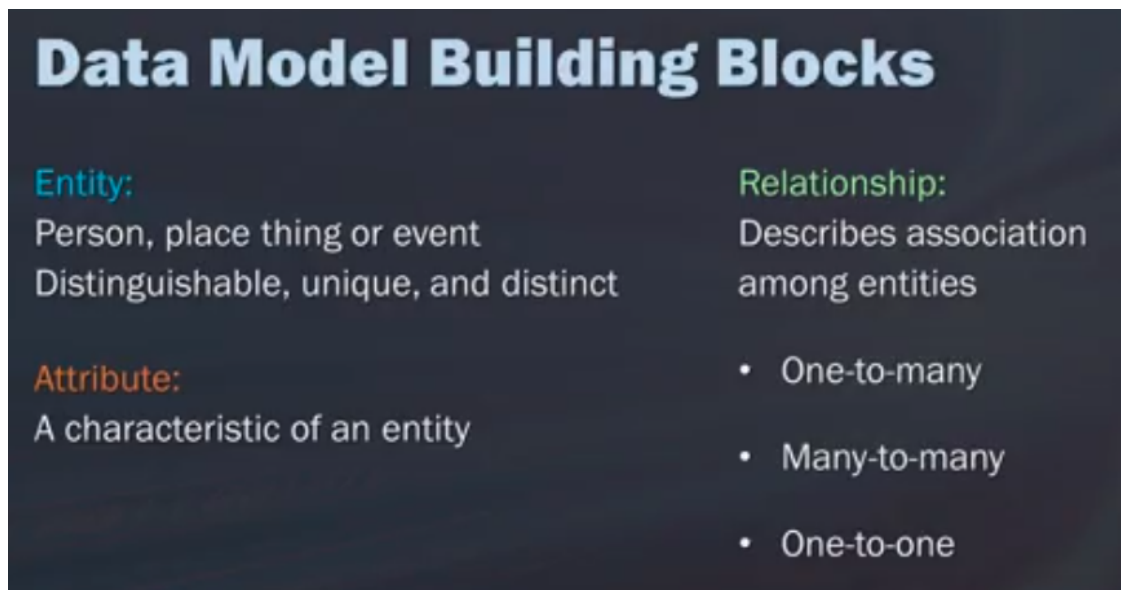
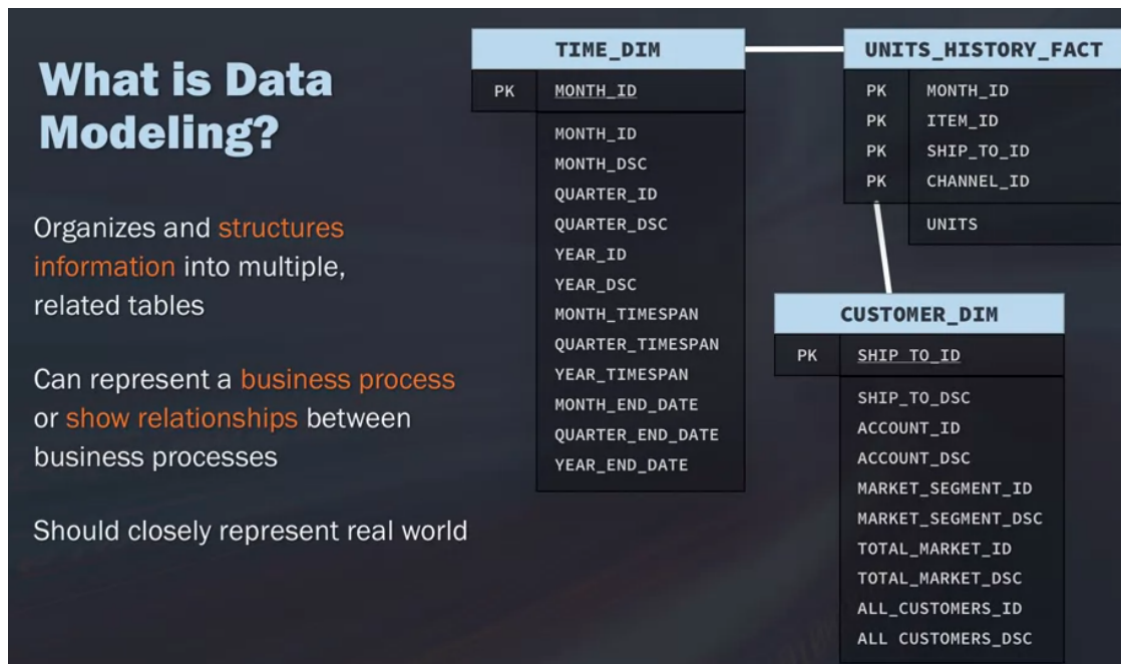


1. Selecting and Retrieving

February 19, 2025

0.0.1 Definitions



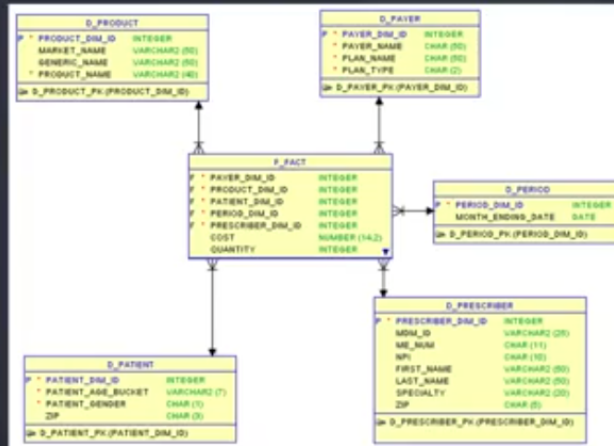
ER Diagrams

Show relationships

Business process

Represented visually

Show links (primary keys)



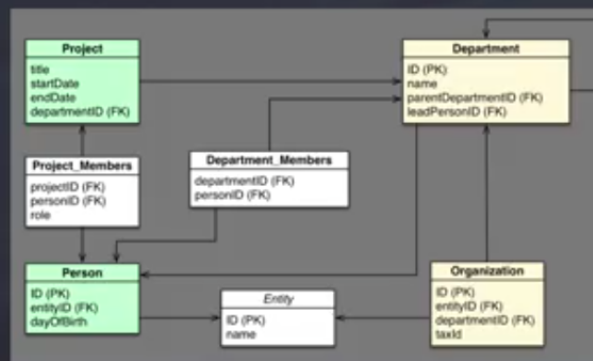
Primary Keys and Foreign Keys

Primary Key

A column (or set of columns) whose values uniquely identify every row in a table

Foreign Key

One or more columns that can be used together to identify a single row in another table



ER Diagram Notation

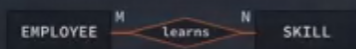
Chen Notation

A One-to-Many (1:M) Relationship: a PAINTER can paint many PAINTINGs; each PAINTING is painted by one PAINTER.

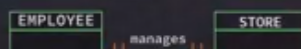


Crow's Foot Notation

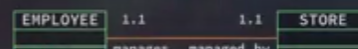
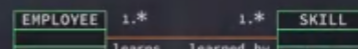
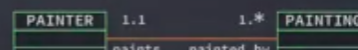
A Many-to-Many (M:N) Relationship: an EMPLOYEE can learn many SKILLs; each SKILL can be learned by many EMPLOYEEs.



A One-to-One (1:1) Relationship: an EMPLOYEE manages one STORE; each STORE is managed by one EMPLOYEE.



UML Class Diagram Notation



0.0.2 Creating Tables

Why Tables are Useful

Use tables to make models and predictions

Create dashboards

Visualize data with other tools

Extract data from other sources

Creating Your Own Table

```
CREATE TABLE Shoes
(
  Id          char(10)      PRIMARY KEY,
  Brand       char(10)      NOT NULL,
  Type        char(250)     NOT NULL,
  Color       char(250)     NOT NULL,
  Price       decimal(8,2)  NOT NULL,
  Desc        Varchar (750) NULL
);
```

Nulls and Primary Keys

Every column is either NULL or NOT NULL

An error will be returned if one tries to submit a column with no value

Don't confuse null values with empty strings

Primary keys can not be null

```
CREATE TABLE Shoes
(
  Id          char(10)      PRIMARY KEY,
  Brand       char(10)      NOT NULL,
  Type        char(250)     NOT NULL,
  Color       char(250)     NOT NULL,
  Price       decimal(8,2)  NOT NULL,
  Desc        Varchar (750) NULL
);
```

Adding Data to the Table

```
INSERT INTO Shoes
VALUES ('14535974'
      , 'Gucci'
      , 'Slippers'
      , 'Pink'
      , '695.00'
      , NULL
      );
```

```
INSERT INTO Shoes
(
  Id
  , Brand
  , Type
  , Color
  , Price
  , Desc
)
VALUES
('14535974'
 , 'Gucci'
 , 'Slippers'
 , 'Pink'
 , '695.00'
 , NULL
);
```

Why Create Temporary Tables?

Temporary tables will be deleted when current session is terminated

Faster than creating a real table

Useful for complex queries using subsets and joins

How to Create a Temporary Table

```
CREATE TEMPORARY TABLE Sandals AS
(
    SELECT *
    FROM shoes
    WHERE shoe_type = 'sandals'
)
```

Adding Comments

Single Line

```
SELECT shoe_id  
- -,brand_id  
,shoe_name  
from shoes
```

Section

```
SELECT shoe_id  
/*,brand_id  
,shoe_name  
*/  
from shoes
```

Source Code Editors

Environment **separate from the database**
where you can write and edit code

e.g.: Notepad++

Automatically highlights and indents
statements

Helps you write clean code

Limiting Results Using Different Syntaxes

SQLite

```
SELECT prod_name  
FROM Products  
LIMIT 5;
```

Oracle

```
SELECT prod_name  
FROM Products  
WHERE ROWNUM <=5;
```

DB2

```
SELECT prod_name  
FROM Products  
FETCH FIRST 5 ROWS ONLY;
```